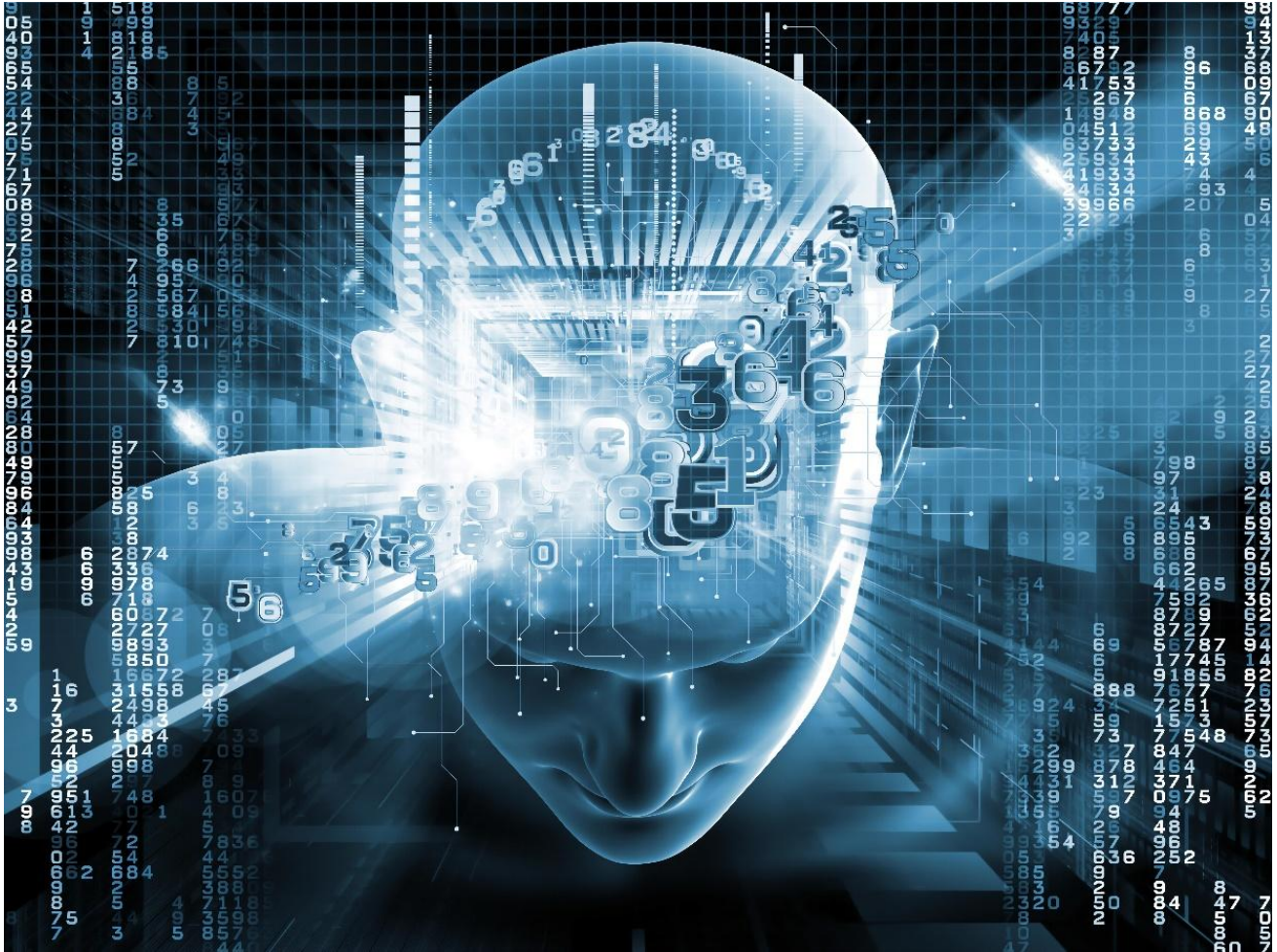




Claude for Finance – Power BI Session 4





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Training Course Documentation

Power BI Acceleration — Session 4 Practical Exercises

Overview

Starting File: Finance-Demo-Model (starting file).xlsx — 2 sheets (Read Me + Fact_Financials with 7,812 rows of monthly financial data)

Solution File: Finance-Demo-Model (final solution).xlsx — 9 sheets (star schema with 5 dimension tables, DAX measures reference, relationships reference)

Data Structure: Star schema — 1 fact table (Fact_Financials) + 5 dimensions (Dim_Date, Dim_Entity, Dim_Account, Dim_Department, Dim_Scenario). Monthly granularity, Jan 2024 – Dec 2026.

Key Infrastructure: 3 entities (Americas, EMEA, APAC), 3 scenarios (Actual, Budget, Forecast), 8 accounts (Revenue / COGS / OpEx), 5 departments. All amounts in AUD. Revenue positive, COGS and expenses negative.

Phase 1: Star Schema Data Preparation

Skill: /pbi-dimension-builder | Sheets Built: Dim_Scenario, Dim_Entity, Dim_Department, Dim_Account, Dim_Date

What Was Built: 5 dimension tables extracted and designed from the Fact_Financials key columns. Each table decodes integer foreign keys into descriptive labels for Power BI slicing and filtering.

Key Design Patterns: Star schema with single fact table. All dimensions connected via integer keys (many-to-one). Dim_Date includes Australian fiscal year (Jul–Jun), quarters, and a MonthIndex for time intelligence. Dim_Account includes AccountGroup, StatementLine, IsExpense, and SortOrder for P&L structuring.

Design Decisions: Sign convention (revenue positive, COGS/expenses negative) eliminates the need for sign-flip logic — Net Profit = SUM(Amount). SortOrder in Dim_Account enables correct P&L line ordering in visuals. Currency standardised to AUD across all entities.

Phase 2: Power BI Reference Sheets & Data Import

Skill: /pbi-reference-builder | Sheets Built: Measures (DAX), Relationships | Power BI Steps: Import 6 tables, create 5 relationships, mark date table

What Was Built

Two reference sheets guide the Power BI setup: Relationships documents the 5 Many-to-One joins from Fact_Financials to each dimension table. Measures (DAX) lists all 14 DAX expressions with syntax and notes. These sheets are not imported into Power BI — they serve as the build specification for Steps 3–5.

Power BI import covers 6 tables (Fact_Financials + 5 Dims). Read Me, Measures, and Relationships are skipped. Column types verified in Power Query: integer keys as Whole Number, Amount as Decimal, MonthEndDate as Date.

Teaching Points

Star schema design separates facts from dimensions. Integer keys enable fast relationships. Marking Dim_Date as a date table unlocks time intelligence functions. The reference sheets bridge the Excel build to the Power BI configuration.



Phase 3: DAX Measures & Calculations

Skill: /pbi-dax-measures | Measures Created: 14 DAX measures across 4 categories

What Was Built

14 DAX measures entered in Power BI, organised into 4 tiers: (1) Core Financial — Total Revenue, Total COGS, Gross Profit, Gross Margin %; (2) Profitability — Total Opex, Operating Profit, Operating Margin %, Net Profit; (3) Scenario & Variance — Actuals, Budget, Variance to Budget, Variance to Budget %; (4) Time Intelligence — Revenue PY, Revenue YoY %.

Key DAX Patterns

CALCULATE with dimension filters: `CALCULATE( SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "Revenue" )`. Measure-to-measure chaining: `[Gross Profit] = [Total Revenue] + [Total COGS]`. Safe division: `DIVIDE([Variance to Budget], ABS([Budget]))`. Time intelligence: `CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[MonthEndDate]))`.

Teaching Points

CALCULATE is the most important DAX function — it modifies filter context. Measures that reference other measures create a calculation chain (Total Revenue → Gross Profit → Operating Profit → Net Profit). The sign convention eliminates complex IF logic — addition works because COGS is already negative. Time intelligence functions require Dim_Date to be marked as a date table.

Phase 4: Dashboard Building & Visualisation

Pages Built: Home (navigation) + P&L Insights – Actual, Budget, Forecast, Variance

What Was Built

Home Page: Navigation hub with 4 buttons linking to each P&L Insights page. Clean layout with report title.

P&L Insights – Actual: Revenue by Entity stacked bars (Americas/APAC/EMEA), Entity slicer, P&L Matrix (AccountGroup rows × FY2024–FY2027 columns), Gross Margin KPI (45.6%), Act vs Budget Revenue KPI (-\$7M), Variance by Entity bars, 4 slicers (Account, AccountGroup, Department, Region). Page-level filter: Scenario = Actual.

P&L Insights – Budget: Same layout as Actual page (duplicated and refiltered). Gross Margin KPI shows 49.5%. Page-level filter: Scenario = Budget. The higher budget margin vs actual immediately signals costs ran above plan.

P&L Insights – Forecast: Same layout, filtered to Scenario = Forecast. Only FY2026–FY2027 have data (FY2024–2025 show blanks — no retrospective forecast). Gross Margin KPI shows 45.8%. Additional KPI: FY27 Forecast vs Budget variance (-\$1.2M).

P&L Insights – Variance: Act vs Budget revenue variance matrix by AccountGroup and FY. 4 Gross Margin KPI cards (Actual 45.6%, Budget 49.5%, Forecast 45.8%, Budget 49.5%). Forecast vs Budget P&L for FY2027 only. This is the executive summary page.



Teaching Points

Summary of Skills Created

1. /pbi-dimension-builder — Builds the 5 star schema dimension tables (Dim_Scenario, Dim_Entity, Dim_Department, Dim_Account, Dim_Date) from any flat fact table with integer keys. Includes Australian fiscal year logic for Dim_Date.\n2. /pbi-reference-builder — Creates Relationships sheet (5 Many-to-One joins) and Measures (DAX) sheet (14 measures covering revenue, margins, scenario filters, variance, and time intelligence). Sign convention notes included.

Workbook Architecture

The starting file contains only Fact_Financials (7,812 rows of monthly financial data with integer keys) and a Read Me sheet. By the end of Prompts 1–2, the workbook has 9 sheets forming a complete star schema: 1 fact table (Fact_Financials), 5 dimension tables (Dim_Date, Dim_Entity, Dim_Account, Dim_Department, Dim_Scenario), and 2 reference sheets (Measures DAX, Relationships). Power BI Steps 1–10 then import, model, measure, and visualise this data as a 5-page interactive report.

Complete Prompt Playbook — Starting File to Solution File

Prompt 1 — Build the Star Schema (Dimension Tables)

Claude Prompt:

"Analyse the Fact_Financials sheet and build 5 dimension tables as new sheets:

1. Dim_Scenario — columns: ScenarioKey, Scenario, ScenarioOrder. Map: 1=Actual (order 1), 2=Budget (order 2), 3=Forecast (order 3).
2. Dim_Entity — columns: EntityKey, Entity, Region, Currency. Map: 1=Americas/North America/AUD, 2=EMEA/Europe & Middle East/AUD, 3=APAC/Asia Pacific/AUD.
3. Dim_Department — columns: DeptKey, Department, DeptGroup. Map: 1=Sales/Commercial, 2=Marketing/Commercial, 3=R&D/Product, 4=Finance/Corporate, 5=Operations/Corporate.
4. Dim_Account — columns: AccountKey, Account, AccountGroup, StatementLine, IsExpense, SortOrder. Map the 8 unique AccountKey values: Product Revenue and Services Revenue (Revenue/Revenue/No/SortOrder 1), Cost of Goods Sold (COGS/Cost of Sales/Yes/2), Salaries & Wages, Marketing & Advertising, Software & Tools, Travel & Entertainment, Other Operating Expenses (all Operating Expenses/Operating Expenses/Yes/3).
5. Dim_Date — columns: DateKey (YYYYMMDD integer using month-end dates), MonthEndDate, Year, Month, MonthName, Quarter, FiscalYear (Australian Jul–Jun, e.g. FY2024 for Jul 2023–Jun 2024), FiscalQuarter (e.g. FQ3 for Jan–Mar), MonthIndex (sequential 1–36). Cover January 2024 to December 2026 (36 rows).

Format each as a proper Excel table with headers in row 1. These tables will connect to Fact_Financials via integer keys in a star schema for Power BI."

Insert validations and checks to make sure that the tables you have built tie back to the Fact Financials



Prompt 2 — Build Reference Sheets (Relationships & DAX Measures)

Claude Prompt:

"Build two Power BI reference sheets:

1. Relationships — columns: From (Fact), Column, To (Dimension), Column, Cardinality, Direction. Document the 5 relationships:

- Fact_Financials.DateKey → Dim_Date.DateKey (Many-to-One, Single)
- Fact_Financials.EntityKey → Dim_Entity.EntityKey (Many-to-One, Single)
- Fact_Financials.AccountKey → Dim_Account.AccountKey (Many-to-One, Single)
- Fact_Financials.DeptKey → Dim_Department.DeptKey (Many-to-One, Single)
- Fact_Financials.ScenarioKey → Dim_Scenario.ScenarioKey (Many-to-One, Single)

2. Measures (DAX) — columns: Measure, DAX Expression, Notes. List 14 DAX measures:

1. Total Revenue = CALCULATE(SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "Revenue")
2. Total COGS = CALCULATE(SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "COGS")
3. Gross Profit = [Total Revenue] + [Total COGS]
4. Gross Margin % = DIVIDE([Gross Profit], [Total Revenue])
5. Total Opex = CALCULATE(SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "Operating Expenses")
6. Operating Profit = [Gross Profit] + [Total Opex]
7. Operating Margin % = DIVIDE([Operating Profit], [Total Revenue])
8. Net Profit = SUM(Fact_Financials[Amount])
9. Actuals = CALCULATE([Net Profit], Dim_Scenario[Scenario] = "Actual")
10. Budget = CALCULATE([Net Profit], Dim_Scenario[Scenario] = "Budget")
11. Variance to Budget = [Actuals] - [Budget]
12. Variance to Budget % = DIVIDE([Variance to Budget], ABS([Budget]))
13. Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[MonthEndDate]))
14. Revenue YoY % = DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY])

Note: COGS and expenses are stored as negative values — addition (not subtraction) produces correct P&L results. These sheets are references only and will not be imported into Power BI."



Prompts for Power BI – Build the instructions to create the Power BI report

Power BI: Import Data

Open Power BI Desktop. File → Get Data → Excel workbook. Navigate to the solution file. Select these 6 tables: Fact_Financials, Dim_Date, Dim_Entity, Dim_Account, Dim_Department, Dim_Scenario. Do NOT import Read Me, Relationships, or Measures (DAX) sheets. In Power Query, verify column types: DateKey as Whole Number, Amount as Decimal Number, MonthEndDate as Date. Remove any blank rows. Click Close & Apply.

Power BI: Create the Data Model

Switch to Model view. Create 5 relationships, all Many-to-One, Single direction, from Fact_Financials to each dimension table: DateKey → Dim_Date.DateKey, EntityKey → Dim_Entity.EntityKey, AccountKey → Dim_Account.AccountKey, DeptKey → Dim_Department.DeptKey, ScenarioKey → Dim_Scenario.ScenarioKey. Then right-click Dim_Date, select Mark as date table, choose MonthEndDate as the date column. This enables time intelligence DAX functions like SAMEPERIODLASTYEAR.

Power BI: Create DAX Measures (14 measures)

Add all measures to the Fact_Financials table:\nRevenue group: Total Revenue = CALCULATE(SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "Revenue"), Total COGS = same pattern with "COGS", Gross Profit = [Total Revenue] + [Total COGS], Gross Margin % = DIVIDE([Gross Profit], [Total Revenue]).\nProfitability group: Total Opex = CALCULATE(SUM(...), "Operating Expenses"), Operating Profit = [Gross Profit] + [Total Opex], Operating Margin % = DIVIDE([Operating Profit], [Total Revenue]), Net Profit = SUM(Fact_Financials[Amount]).\nScenario group: Actuals = CALCULATE([Net Profit], Dim_Scenario[Scenario] = "Actual"), Budget = same with "Budget", Variance to Budget = [Actuals] - [Budget], Variance to Budget % = DIVIDE([Variance to Budget], ABS([Budget])).\nTime Intelligence: Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[MonthEndDate])), Revenue YoY % = DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY]).

Build the Home Page

Create a new report page named 'Home'. Add a large title text box: 'Finance Demo Model — Power BI Report'. Create 4 navigation buttons (Insert → Buttons → Navigator) linking to: P&L Insights – Actual, P&L Insights – Budget, P&L Insights – Forecast, P&L Insights – Variance. Set each button's Action to Page navigation. Style with a consistent colour scheme, arrange in a 2x2 grid. Move this page to position 1.

Build P&L Insights – Actual

Create a page named 'P&L Insights – Actual'. Layout: Top-left: stacked bar chart showing Total Revenue by Entity (Americas, APAC, EMEA) across FY2024–FY2027. Top-right: Entity slicer (default All). Centre: Matrix visual with AccountGroup on rows (Revenue, COGS, Operating Expenses), FiscalYear columns (FY2024, FY2025, FY2026, FY2027, Total). Values = Actuals measure. Bottom-left: KPI card showing Act vs Budget Revenue variance (approximately -\$7M). Bottom-right: Gross Margin % KPI card (45.6%). Add 4 slicers: Account, AccountGroup, Department, Region. Add a Clustered Bar Chart showing Variance to Budget by Entity. Pre-filter the page to Scenario = Actual using a page-level filter. Add a Back button linking to the Home page.



Build P&L Insights – Budget

Duplicate the P&L Insights – Actual page. Rename to 'P&L Insights – Budget'. Change the page-level filter to Dim_Scenario[Scenario] = Budget. Update the P&L Matrix values to use the Budget measure. Update the Gross Margin % KPI — it should show 49.5% (Budget gross margin is higher than Actual). Update the variance KPI and bar chart for Budget context. Keep the same layout, slicers, and entity breakdown.

Build P&L Insights – Forecast

Duplicate the P&L Insights – Actual page. Rename to 'P&L Insights – Forecast'. Change the page-level filter to Dim_Scenario[Scenario] = Forecast. Note: Forecast data only covers FY2026–FY2027, so FY2024 and FY2025 columns will show blanks — this is correct. Update the Gross Margin % KPI to 45.8%. Add a new KPI card: FY27 Forecast vs Budget Revenue variance (approximately -\$1.2M). The stacked bar chart will naturally show only FY2026–FY2027 bars.

Build P&L Insights – Variance

Create a new page (do not duplicate) named 'P&L Insights – Variance'. Top: Act vs Budget Revenue Variance matrix showing Revenue lines (Product Revenue, Services Revenue) by FY2024–FY2027 + Total. All values in parentheses (negative = unfavourable). KPI card: Act vs Budget Revenue -\$7.2M. Middle: 4 Gross Margin KPI cards side by side — Actual Gross Margin % (45.6%), Budget Gross Margin % (49.5%), Forecast Gross Margin % (45.8%), Budget Gross Margin % (49.5%). Bottom: Forecast vs Budget Profit and Loss matrix showing FY2027 only. KPI: FY27 Fcst vs Budget Revenue (-\$1.2M). Add entity slicer and 4 additional slicers (Account, AccountGroup, Department, Region).



Power BI Steps — From Excel to Interactive Dashboards

These 10 steps are performed manually in Power BI Desktop — they are not Claude prompts. They follow directly from the Excel workbook built in Prompts 1–2. The endpoint is a 5-page report: Home (navigation) + 4 P&L Insights pages (Actual, Budget, Forecast, Variance).

Step 1 — Get Data: Connect to Excel

1. Open Power BI Desktop → Get Data → Excel Workbook → navigate to Finance-Demo-Model (final solution).xlsx.\n2. In the Navigator pane, tick these 6 tables: Fact_Financials, Dim_Date, Dim_Entity, Dim_Account, Dim_Department, Dim_Scenario.\n3. Do NOT import: Read Me (documentation only), Measures (DAX) (reference only — you'll type these as PBI measures), Relationships (reference only — you'll create these in Model view).\n4. Click Load (or Transform Data if you want to verify column types first).

Step 2 — Transform Data: Verify Column Types

1. If you clicked Transform Data, Power Query Editor opens. Check each table:\n • Fact_Financials: DateKey (Whole Number), EntityKey (Whole Number), AccountKey (Whole Number), DeptKey (Whole Number), ScenarioKey (Whole Number), Amount (Decimal Number).\n • Dim_Date: DateKey (Whole Number), MonthEndDate (Date), Year/Month/MonthIndex (Whole Number), all text columns (Text).\n • All other Dim_ tables: Key columns (Whole Number), label columns (Text).\n2. Fix any misdetected types (e.g. if DateKey loaded as Text, change to Whole Number).\n3. Click Close & Apply to load all tables into the data model.

Step 3 — Create Relationships in Model View

1. Switch to Model view (left sidebar icon).\n2. You should see all 6 tables as boxes. If Power BI auto-detected relationships, verify they match the Relationships reference sheet. If not, create them manually:\n • Drag Fact_Financials[DateKey] → Dim_Date[DateKey]\n • Drag Fact_Financials[EntityKey] → Dim_Entity[EntityKey]\n • Drag Fact_Financials[AccountKey] → Dim_Account[AccountKey]\n • Drag Fact_Financials[DeptKey] → Dim_Department[DeptKey]\n • Drag Fact_Financials[ScenarioKey] → Dim_Scenario[ScenarioKey]\n3. Verify each relationship shows: Many-to-One (many side on Fact_Financials), Single filter direction (arrow pointing from Fact to Dim).\n4. The star schema should be visible — Fact_Financials in the centre with 5 dimension tables radiating outward.

Step 4 — Mark Date Table

1. In Model view, click the Dim_Date table.\n2. Go to Table tools ribbon → Mark as Date Table → select MonthEndDate as the date column.\n3. This is essential — without it, time intelligence functions (SAMEPERIODLASTYEAR, DATEADD) will not work.\n4. Power BI will validate that MonthEndDate contains unique, contiguous date values with no gaps.

Step 5 — Create All 14 DAX Measures

1. Switch to Report view. In the Fields pane, right-click Fact_Financials → New Measure.
2. Enter all 14 measures from the Measures (DAX) reference sheet:

Revenue & Margins (8 measures):

- Total Revenue = CALCULATE(SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "Revenue")
- Total COGS = CALCULATE(SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "COGS")



MODEL CITIZEN

- Gross Profit = [Total Revenue] - [Total COGS]
- Gross Margin % = DIVIDE([Gross Profit], [Total Revenue])
- Total Opex = CALCULATE(SUM(Fact_Financials[Amount]), Dim_Account[AccountGroup] = "Operating Expenses")
- Operating Profit = [Gross Profit] - [Total Opex]
- Operating Margin % = DIVIDE([Operating Profit], [Total Revenue])
- Net Profit = SUM(Fact_Financials[Amount])

Scenario & Variance (4 measures):

- Actuals = CALCULATE([Net Profit], Dim_Scenario[Scenario] = "Actual")
- Budget = CALCULATE([Net Profit], Dim_Scenario[Scenario] = "Budget")
- Variance to Budget = [Actuals] - [Budget]
- Variance to Budget % = DIVIDE([Variance to Budget], ABS([Budget]))

Time Intelligence (2 measures):

- Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[MonthEndDate]))
- Revenue YoY % = DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY])

3. Format percentage measures as % (Measure tools → Format → Percentage).

4. Verify: add a test Table visual with Dim_Date[MonthName] and all measures. Revenue should be positive, COGS negative, margins 0–100%.

Teaching Point: CALCULATE changes filter context — it's the most important DAX function. SAMEPERIODLASTYEAR only works because you marked Dim_Date as a date table in Step 4.



Step 6 — Build Home Page

1. Add a new blank page and rename the tab to 'Home'.
2. Add a large title text box: 'Finance Demo Model — Power BI Report'.
3. Create 4 navigation buttons (Insert → Buttons → Navigator). Label each:
 - P&L Insights – Actual
 - P&L Insights – Budget
 - P&L Insights – Forecast
 - P&L Insights – Variance
4. Set each button's Action to navigate to the corresponding page (Format → Action → Page navigation).
5. Style the buttons with a consistent colour scheme. Arrange in a 2×2 grid centred on the page.
6. Move this page to position 1 (drag the tab to the left of all other pages).

Teaching Point: A Home page with navigation buttons creates a professional user experience — users don't need to scroll through page tabs.

Step 7 — Build P&L Insights – Actual

1. Add a new page, rename to 'P&L Insights – Actual'.
2. Add a Stacked Bar Chart: Axis = Dim_Date[FiscalYear], Values = Total Revenue, Legend = Dim_Entity[Entity]. This shows revenue contribution by entity (Americas, APAC, EMEA) across FY2024–FY2027.
3. Add an Entity Slicer: Dim_Entity[Entity] — enables filtering the entire page by entity.
4. Add a Matrix visual: Rows = Dim_Account[AccountGroup], Columns = Dim_Date[FiscalYear] (FY2024, FY2025, FY2026, FY2027, plus Grand Total). Values = Actuals. This is the core P&L matrix showing Revenue, COGS, and Operating Expenses by year. Sort rows using column tools

AccountKey	AccountName	AccountGroup	StatementLine	IsExpense	SortOrder
1	Revenue	Revenue	Total Revenue	No	1
2	Cost of Goods Sold	COGS	Total COGS	Yes	2
3	Salaries & Wages	Operating Expenses	Total Operating Expenses	Yes	3
4	Rent & Occupancy	Operating Expenses	Total Operating Expenses	Yes	4
5	Marketing Expense	Operating Expenses	Total Operating Expenses	Yes	5
6	Depreciation	Operating Expenses	Total Operating Expenses	Yes	6
7	Interest Expense	Finance Costs	Total Finance Costs	Yes	7
8	Other Income	Other Income	Total Other Income	No	8

5. Add a KPI Card for Gross Margin %: value should show 45.6% (the Actual gross margin across all periods).
6. Add a KPI Card for Act vs Budget Revenue: showing the total variance (approximately -\$7M).



7. Add a Clustered Bar Chart: Axis = Dim_Entity[Entity], Values = Variance to Budget. Shows which entities are over/under budget.
8. Add 4 Slicers in a row: Dim_Account[Account], Dim_Account[AccountGroup], Dim_Department[Department], Dim_Entity[Region].
9. Pre-filter the page to Scenario = Actual using a page-level filter (Filters pane → Dim_Scenario[Scenario] = Actual).
10. Add a Back button linking to the Home page.

Teaching Point: Page-level filters lock the scenario context — users interact with entity, department, and account slicers without accidentally switching scenarios.



Step 8 — Build P&L Insights – Budget

1. Duplicate the P&L Insights – Actual page (right-click tab → Duplicate Page).
2. Rename the tab to 'P&L Insights – Budget'.
3. Change the page-level filter: Dim_Scenario[Scenario] = Budget.
4. Update the P&L Matrix values to use the Budget measure instead of Actuals.
5. Update the Gross Margin % KPI — it should now show 49.5% (Budget gross margin is higher than Actual).
6. Update the variance KPI and bar chart to show Budget vs Forecast variance (if meaningful) or remove if not applicable.
7. Update the page title to indicate Budget view.
8. Verify all slicers and cross-filtering still work correctly.

Teaching Point: Duplicating a page and changing the scenario filter is far faster than rebuilding from scratch. The Budget gross margin of 49.5% vs Actual 45.6% immediately tells the story — costs ran higher than planned.

Step 9 — Build P&L Insights – Forecast

1. Duplicate the P&L Insights – Actual page again.
 2. Rename the tab to 'P&L Insights – Forecast'.
 3. Change the page-level filter: Dim_Scenario[Scenario] = Forecast.
 4. Update the P&L Matrix to use Forecast data. Note: Forecast data only covers FY2026–FY2027, so FY2024 and FY2025 columns will show blanks — this is correct (no retrospective forecast exists).
 5. Update the Gross Margin % KPI — it should show 45.8% (Forecast margin).
 6. Add a new KPI Card: FY27 Forecast vs Budget variance, showing approximately -\$1.2M — the forecast expects to come in slightly under budget.
 7. The stacked bar chart will naturally show only FY2026–FY2027 bars (no forecast data for prior years).
 8. Update page title and verify all slicers work.
- Teaching Point: Forecast data being forward-looking means historical columns are intentionally blank. This is a feature, not a bug — users immediately see the forecast horizon.

Step 10 — Build P&L Insights – Variance

1. Add a new page (do not duplicate — this page has a different layout), rename to 'P&L Insights – Variance'.
 2. Add a Matrix visual: Rows = Dim_Account[AccountGroup], Columns = Dim_Date[FiscalYear]. Values = Variance to Budget. This shows Actual vs Budget revenue variance by account group and year.
 3. Add 4 KPI Cards in a row showing Gross Margin % for each scenario perspective:
 - Actual Gross Margin: 45.6%
 - Budget Gross Margin: 49.5%
 - Forecast Gross Margin: 45.8%
 - Budget Gross Margin (repeated for comparison): 49.5%
 4. Add a Forecast vs Budget P&L section: a Matrix with Rows = Dim_Account[AccountGroup], filtered to FY2027 only, showing Forecast and Budget values side by side with a Variance column.
 5. Add entity and account slicers consistent with other pages.
 6. Add a Back button linking to the Home page.
- Teaching Point: The Variance page is the executive summary of the entire report — it answers "how are we tracking against plan?" across all dimensions in one view. The 4 margin KPIs give instant context without clicking.



Short Form Prompts

Prompt S1 — Excel Data Prep (Condensed)

Claude Prompt:

"Build 5 dimension tables from Fact_Financials (Dim_Scenario: 3 scenarios with keys/order; Dim_Entity: 3 entities with region/currency; Dim_Department: 5 depts with groups; Dim_Account: 8 accounts with AccountGroup/StatementLine/IsExpense/SortOrder; Dim_Date: 36-month calendar Jan 2024–Dec 2026 with Australian fiscal year Jul–Jun). Then build 2 reference sheets: Relationships (5 Many-to-One joins from Fact to each Dim) and Measures DAX (14 measures covering revenue, margins, scenario filters, variance, time intelligence). Format all as tables. Insert validations and checks to make sure that the tables you have built tie back to the Fact Financials."

Prompt S2 — Power BI Setup (Condensed) - new

"Prepare instructions on a new sheet for me to follow based on the following steps:

Power BI Steps:\n1. Get Data → Excel → import 6 tables (Fact_Financials + 5 Dims). Skip Read Me, Measures, Relationships.\n2. Transform Data → verify DateKey as Whole Number, Amount as Decimal Number, MonthEndDate as Date.\n3. Model View → drag 5 relationships: Fact_Financials keys to each Dim table primary key (Many-to-One, Single direction).\n4. Right-click Dim_Date → Mark as date table → select MonthEndDate as the date column."

Prompt S3 — DAX Measures (Condensed)

"Prepare instructions on a new sheet for me to follow based on the following steps:

Create 14 DAX measures in Fact_Financials:\nRevenue & Margins: Total Revenue (CALCULATE+SUM, AccountGroup="Revenue"), Total COGS (AccountGroup="COGS"), Gross Profit (Revenue+COGS), Gross Margin % (DIVIDE), Total Opex (AccountGroup="Operating Expenses"), Operating Profit (GP+Opex), Operating Margin % (DIVIDE), Net Profit (SUM all).\nScenario & Variance: Actuals (CALCULATE, Scenario="Actual"), Budget (Scenario="Budget"), Variance to Budget (Actuals-Budget), Variance % (DIVIDE with ABS).\nTime Intelligence: Revenue PY (SAMEPERIODLASTYEAR on MonthEndDate), Revenue YoY % (DIVIDE). Format % measures as percentage."

Prompt S4 — Build the Report (Condensed)

"Prepare instructions on a new sheet for me to follow based on the following steps:

Build a 5-page Power BI report:\n1. Home — navigation buttons linking to 4 P&L Insights pages. Clean layout with report title.\n2. P&L Insights – Actual — Revenue by Entity stacked bars, Entity slicer, P&L Matrix (AccountGroup rows × FY columns), Gross Margin KPI (45.6%), Act vs Budget Revenue KPI (-\$7M), Variance by Entity bars, 4 slicers (Account, AccountGroup, Department, Region). Page filter: Scenario = Actual.\n3. P&L Insights – Budget — Same layout, Budget data. Gross Margin 49.5%. Page filter: Scenario = Budget.\n4. P&L Insights – Forecast — Same layout, FY2026–FY2027 only (FY24–25 blank). Gross Margin 45.8%, FY27 Fcst vs Budget KPI (-\$1.2M). Page filter: Scenario = Forecast.\n5. P&L Insights – Variance — Act vs Budget variance matrix, 4 Gross Margin KPIs (Act 45.6%, Budget 49.5%, Fcst 45.8%, Budget 49.5%), Forecast vs Budget P&L (FY2027 only)."